

<https://chatgpt.com/c/69f6d94f-b36c-83a3-8be8-b69d0338c0bb>

Dimensionality Reduction

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Machine Learning 10-701/15-781
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Slides Courtesy: Tom Mitchell, Eric Xing, Lawrence Saul



MACHINE LEARNING DEPARTMENT



High-Dimensional data

- High-Dimensions = Lot of Features

Document classification

Features per document =
thousands of words/unigrams
millions of bigrams, contextual
information



Surveys - Netflix

480189 users x 17770 movies

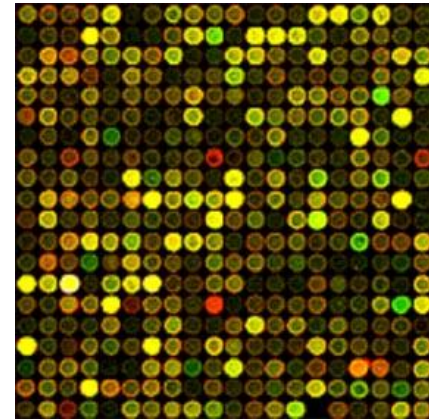
	movie 1	movie 2	movie 3	movie 4	movie 5	movie 6
Tom	5	?	?	1	3	?
George	?	?	3	1	2	5
Susan	4	3	1	?	5	1
Beth	4	3	?	2	4	2

High-Dimensional data

- High-Dimensions = Lot of Features

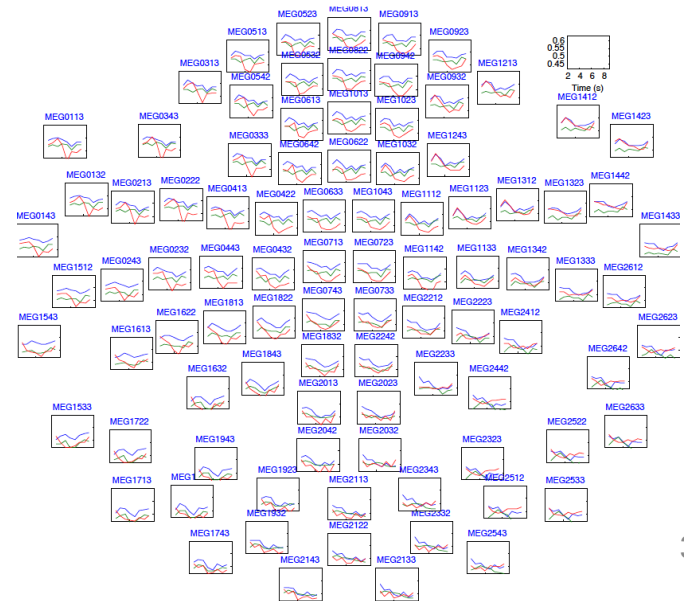
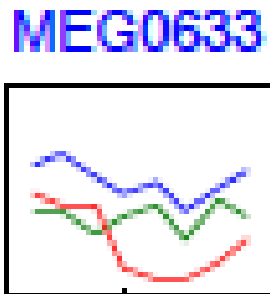
Discovering gene networks

10,000 genes x 1000 drugs
x several species



MEG Brain Imaging

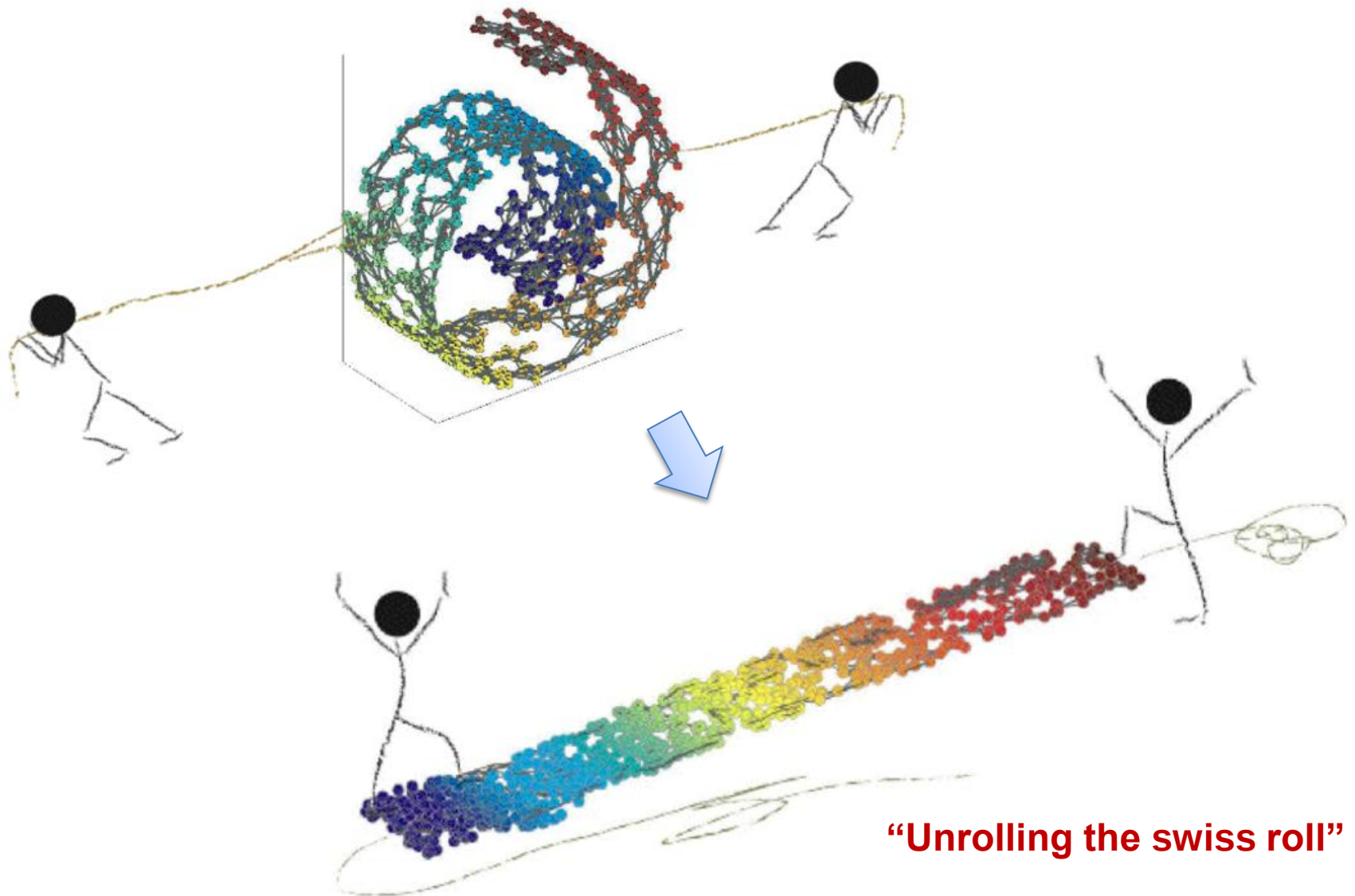
120 locations x 500 time points
x 20 objects



Curse of Dimensionality

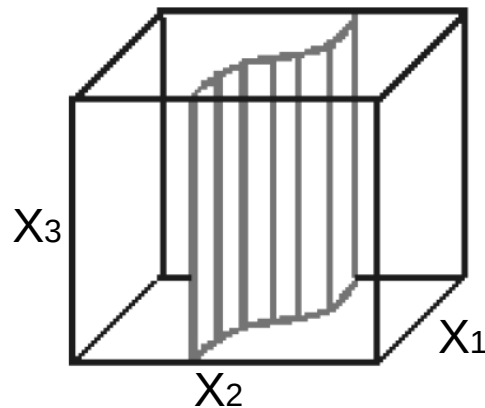
- Why are more features bad?
 - Redundant features (not all words are useful to classify a document)
more noise added than signal
 - Hard to interpret and visualize
 - Hard to store and process data (computationally challenging)
 - Complexity of decision rule tends to grow with # features. Hard to learn complex rules as VC dimension increases (statistically challenging)

Dimensionality Reduction



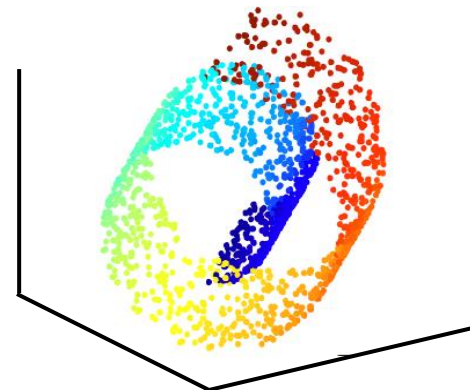
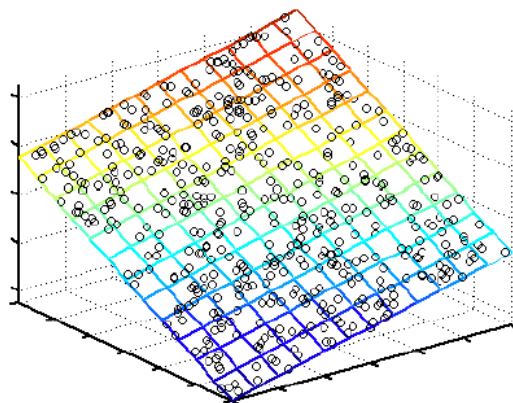
Dimensionality Reduction

- **Feature Selection** – Only a few features are relevant to the learning task



X_3 - Irrelevant

- **Latent features** – Some linear/nonlinear combination of features provides a more efficient representation than observed features



Feature Selection

- Approach 1: Score each feature and extract a subset

Common scoring methods:

- Training or cross-validated accuracy of single-feature classifiers $f_i: X_i \rightarrow Y$
- Estimated mutual information between X_i and Y :
$$\hat{I}(X_i, Y) = \sum_k \sum_y \hat{P}(X_i = k, Y = y) \log \frac{\hat{P}(X_i = k, Y = y)}{\hat{P}(X_i = k) \hat{P}(Y = y)}$$
- χ^2 statistic to measure independence between X_i and Y
- Domain specific criteria
 - Text: Score “stop” words (“the”, “of”, ...) as zero
 - fMRI: Score voxel by T-test for activation versus rest condition
 - ...

Feature Selection

- Approach 1: **Score each feature and extract a subset**

Common subset selection methods:

- One step: Choose d highest scoring features
- Iterative:
 - Choose single highest scoring feature X_k
 - Rescore all features, conditioned on the set of already-selected features
 - E.g., $\text{Score}(X_i | X_k) = I(X_i, Y | X_k)$
 - E.g., $\text{Score}(X_i | X_k) = \text{Accuracy}(\text{predicting } Y \text{ from } X_i \text{ and } X_k)$
 - Repeat, calculating new scores on each iteration, conditioning on set of selected features

Feature Selection: Text Classification

Approximately 10^5 words in English

[Rogati&Yang, 2002]

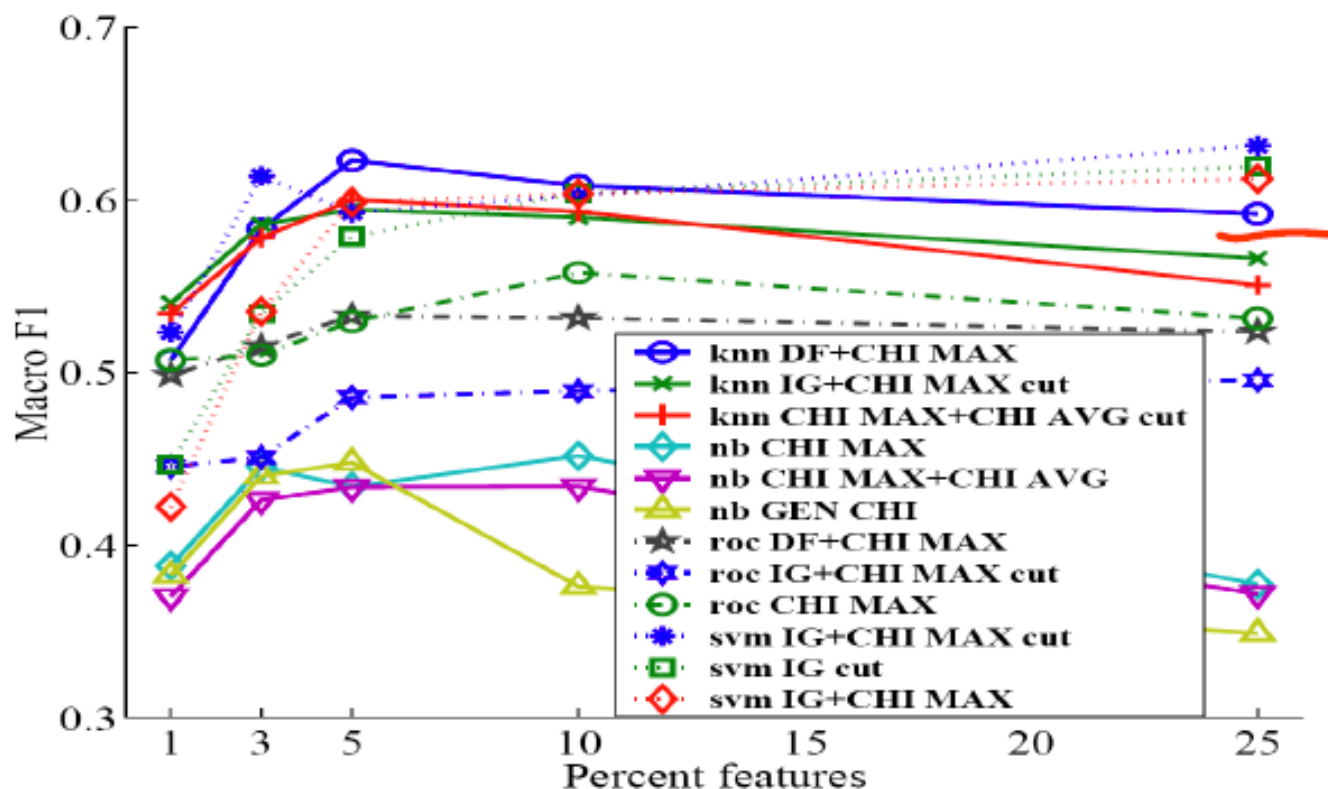


Figure 2: Top 3 feature selection methods for Reuters-21578 (Macro F1)

IG=information gain, chi= χ^2 , DF=doc frequency,

Impact of Feature Selection on Classification of fMRI Data

[Pereira et al., 2005]

Accuracy classifying
category of word read
by subject

#voxels	mean	subjects							
		233B	329B	332B	424B	474B	496B	77B	86B
50	0.735	0.783	0.817	0.55	0.783	0.75	0.8	0.65	0.75
100	0.742	0.767	0.8	0.533	0.817	0.85	0.783	0.6	0.783
200	0.737	0.783	0.783	0.517	0.817	0.883	0.75	0.583	0.783
300	0.75	0.8	0.817	0.567	0.833	0.883	0.75	0.583	0.767
400	0.742	0.8	0.783	0.583	0.85	0.833	0.75	0.583	0.75
800	0.735	0.833	0.817	0.567	0.833	0.833	0.7	0.55	0.75
1600	0.698	0.8	0.817	0.45	0.783	0.833	0.633	0.5	0.75
all (~2500)	0.638	0.767	0.767	0.25	0.75	0.833	0.567	0.433	0.733

Table 1: **Average accuracy across all pairs of categories, restricting the procedure to use a certain number of voxels for each subject.** The highlighted line corresponds to the best mean accuracy, obtained using 300 voxels.

Each feature X_i is a voxel, scored by error in regression to predict X_i from Y

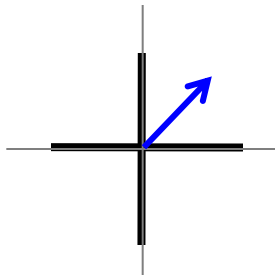
Feature Selection

- Approach 2: **Regularization (MAP)**

Integrate feature selection into learning objective by penalizing number of features with non-zero weights

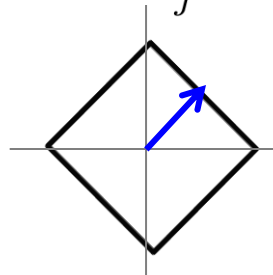
$$\widehat{W} = \arg \min_W \underbrace{\sum_{i=1}^n -\log P(Y_i|X_i; W)}_{\text{-ve log likelihood}} + \underbrace{\lambda \|W\|}_{\text{penalty}}$$

$$\|W\|_0 = \#\{W_j > 0\}$$



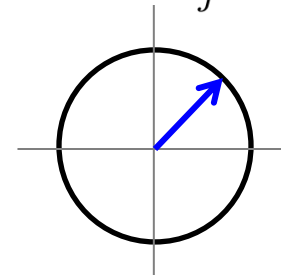
Minimizes # features
chosen

$$\|W\|_1 = \sum_j |W_j|$$



Convex
compromise

$$\|W\|_2 = \sum_j W_j^2$$



Small weights of
features chosen

Latent Feature Extraction

Combinations of observed features provide more efficient representation, and capture underlying relations that govern the data

E.g. Ego, personality and intelligence are hidden attributes that characterize human behavior instead of survey questions

Topics (sports, science, news, etc.) instead of documents

Often may not have physical meaning

- Linear

- Principal Component Analysis (PCA)**

- Factor Analysis

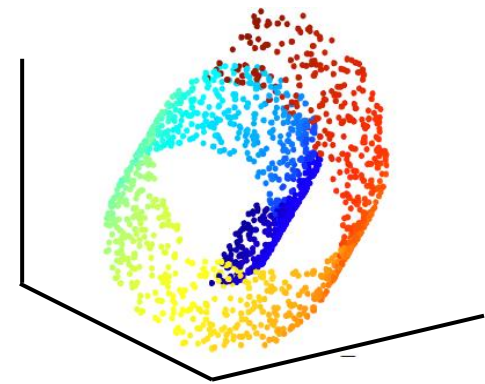
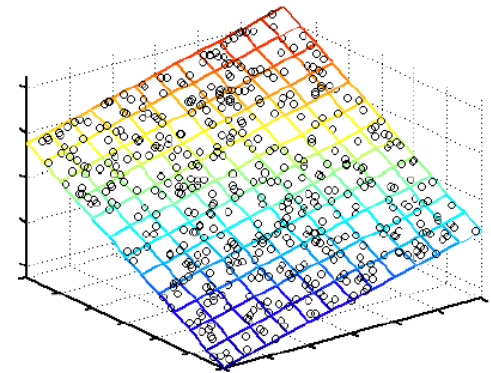
- Independent Component Analysis (ICA)

- Nonlinear

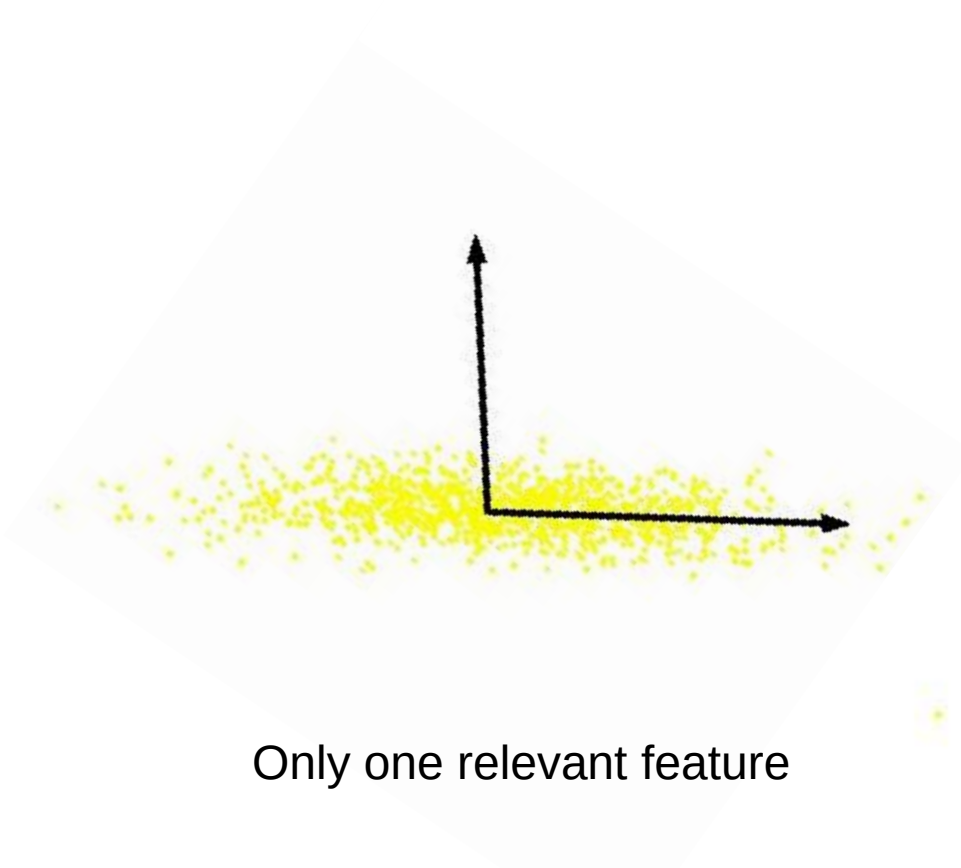
- Laplacian Eigenmaps**

- ISOMAP

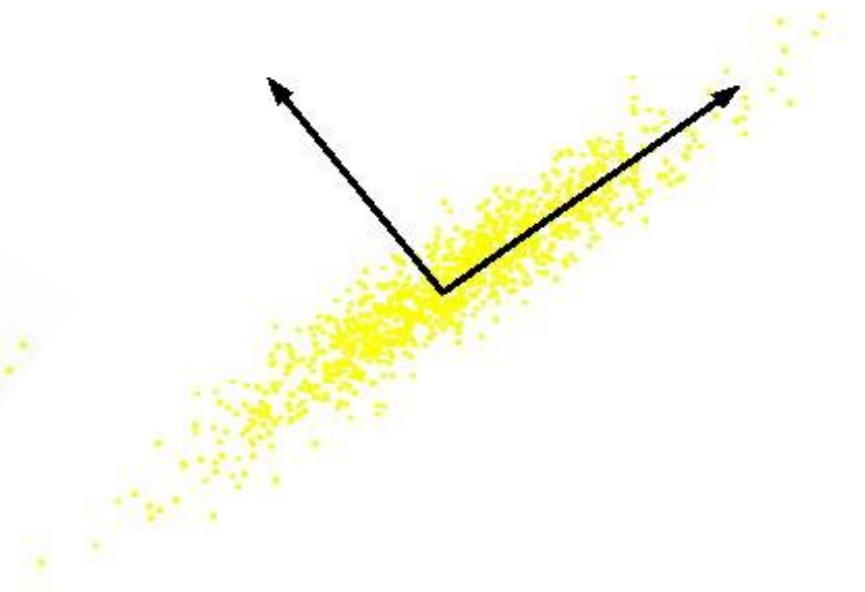
- Local Linear Embedding (LLE)



Principal Component Analysis (PCA)



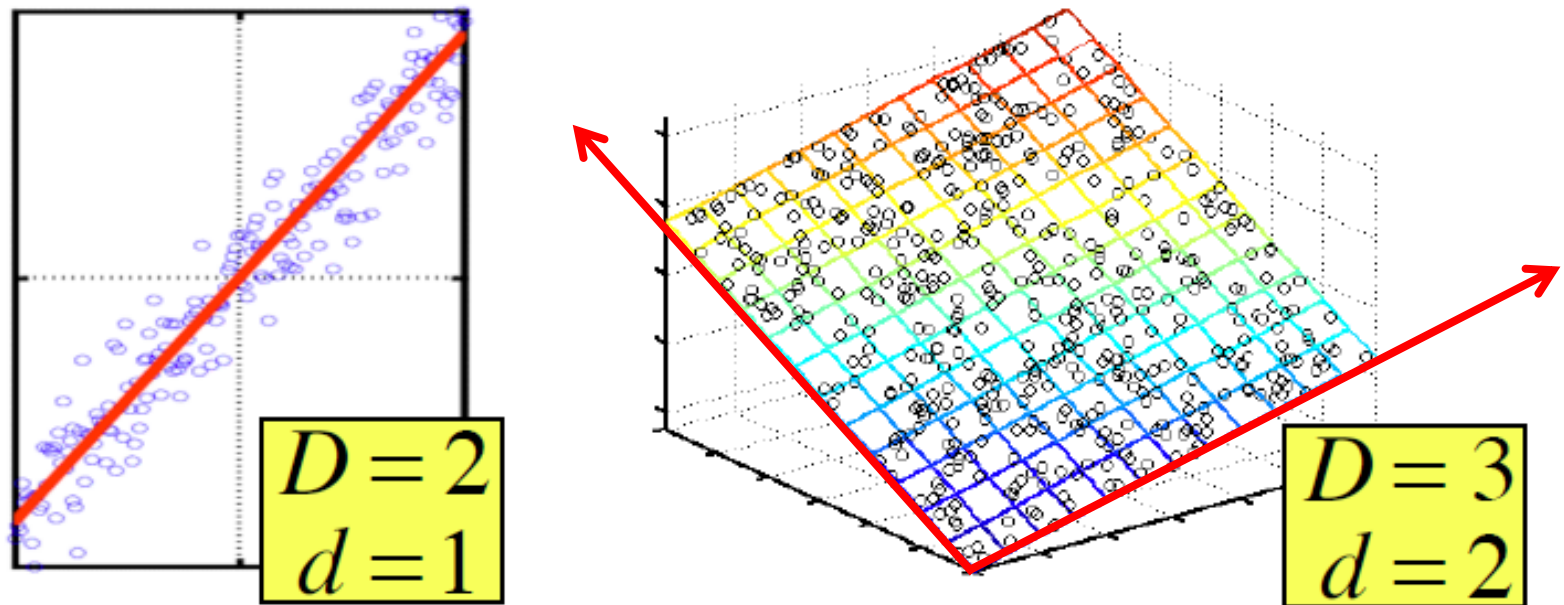
Only one relevant feature



Both features become relevant

Can we transform the features so that we only need to preserve one latent feature? Find linear projection so that projected data is uncorrelated.

Principal Component Analysis (PCA)

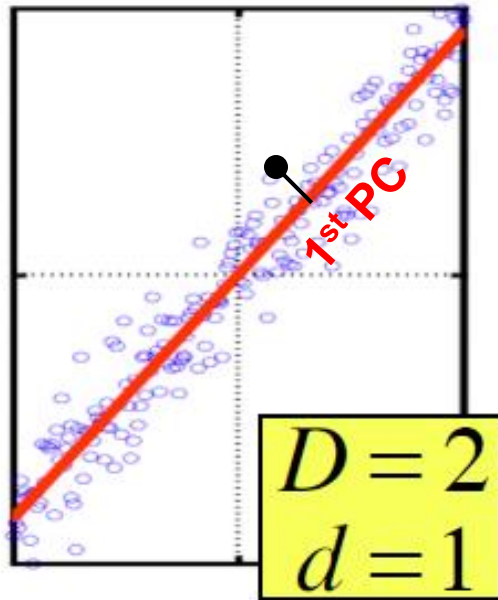


Assumption: Data lies on or near a low d -dimensional linear subspace.

Axes of this subspace are an effective representation of the data

Identifying the axes is known as Principal Components Analysis, and can be obtained by Eigen or Singular value decomposition

Principal Component Analysis (PCA)



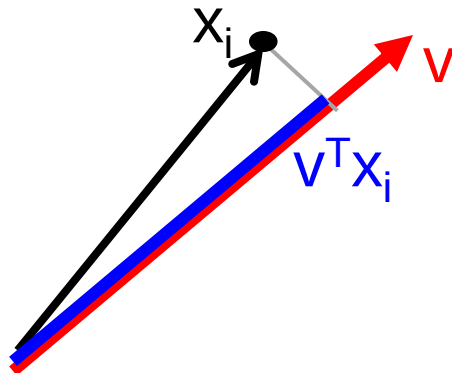
Principal Components (PC) are orthogonal directions that capture most of the variance in the data

1st PC – direction of greatest variability in data

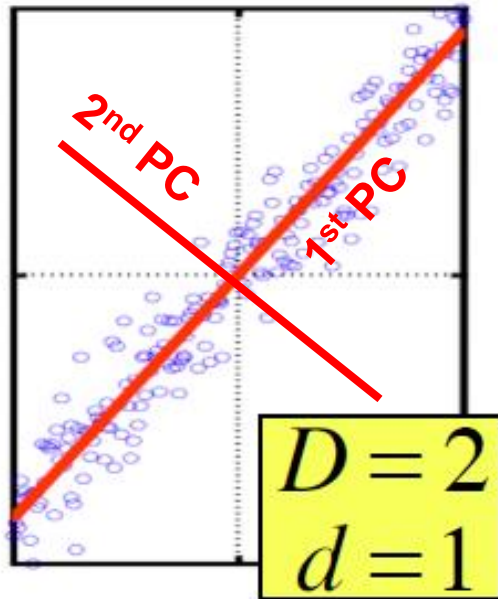
Projection of data points along 1st PC discriminate the data most along any one direction

Take a data point x_i (D -dimensional vector)

Projection of x_i onto the 1st PC v is $v^T x_i$



Principal Component Analysis (PCA)



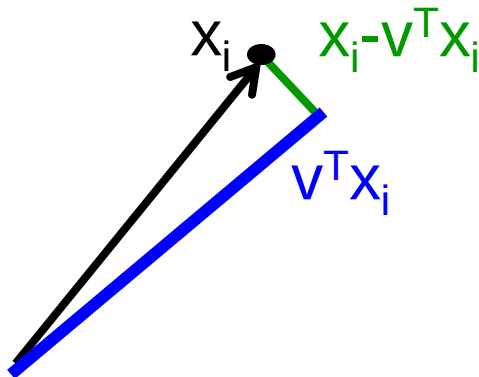
Principal Components (PC) are orthogonal directions that capture most of the variance in the data

1st PC – direction of greatest variability in data

2nd PC – Next orthogonal (uncorrelated) direction of greatest variability

(remove all variability in first direction, then find next direction of greatest variability)

And so on ...



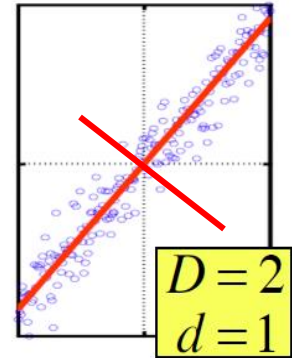
Principal Component Analysis (PCA)

Let $v_1, v_2, \dots, v_d$ denote the principal components

Orthogonal and unit norm $v_i^T v_j = 0 \quad i \neq j$

$$v_i^T v_i = 1$$

Find vector that maximizes sample variance of projection



$$\frac{1}{n} \sum_{i=1}^n (v^T x_i)^2 = v^T X X^T v$$

Assume data are centered
Data points $X = [x_1 \ x_2 \ \dots \ x_n]$

$$\max_v v^T X X^T v \quad \text{s.t.} \quad v^T v = 1$$

Lagrangian: $\max_v v^T X X^T v - \lambda v^T v$

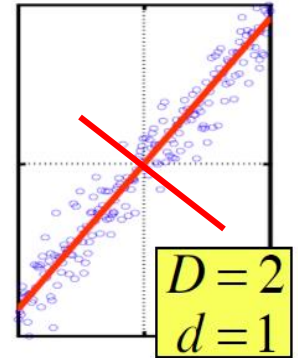
Wrap constraints into the objective function

$$\partial/\partial v = 0 \quad (X X^T - \lambda I)v = 0 \quad \Rightarrow \boxed{(X X^T)v = \lambda v}$$

Principal Component Analysis (PCA)

$$(\mathbf{X}\mathbf{X}^T)\mathbf{v} = \lambda\mathbf{v}$$

Therefore, $\mathbf{v}$ is the eigenvector of sample correlation/covariance matrix $\mathbf{X}\mathbf{X}^T$



Sample variance of **projection** $= \mathbf{v}^T \mathbf{X}\mathbf{X}^T \mathbf{v} = \lambda \mathbf{v}^T \mathbf{v} = \lambda$

Thus, the eigenvalue λ denotes the amount of variability captured along that dimension (aka amount of energy along that dimension).

Eigenvalues $\lambda_1 > \lambda_2 > \lambda_3 > \dots$

The 1st Principal component $\mathbf{v}_1$ is the eigenvector of the sample covariance matrix $\mathbf{X}\mathbf{X}^T$ associated with the largest eigenvalue λ_1

The 2nd Principal component $\mathbf{v}_2$ is the eigenvector of the sample covariance matrix $\mathbf{X}\mathbf{X}^T$ associated with the second largest eigenvalue λ_2

And so on ...

Computing the PCs

Eigenvectors are solutions of the following equation:

$$(\mathbf{X}\mathbf{X}^T)\mathbf{v} = \lambda\mathbf{v} \quad (\mathbf{X}\mathbf{X}^T - \lambda\mathbf{I})\mathbf{v} = 0$$

Non-zero solution $\mathbf{v} \neq 0$ possible only if

$$\det(\mathbf{X}\mathbf{X}^T - \lambda\mathbf{I}) = 0 \quad \text{Characteristic Equation}$$

This is a D^{th} order equation in λ , can have at most D distinct solutions (roots of the characteristic equation)

Once eigenvalues are computed, solve for eigenvectors (Principal Components) using

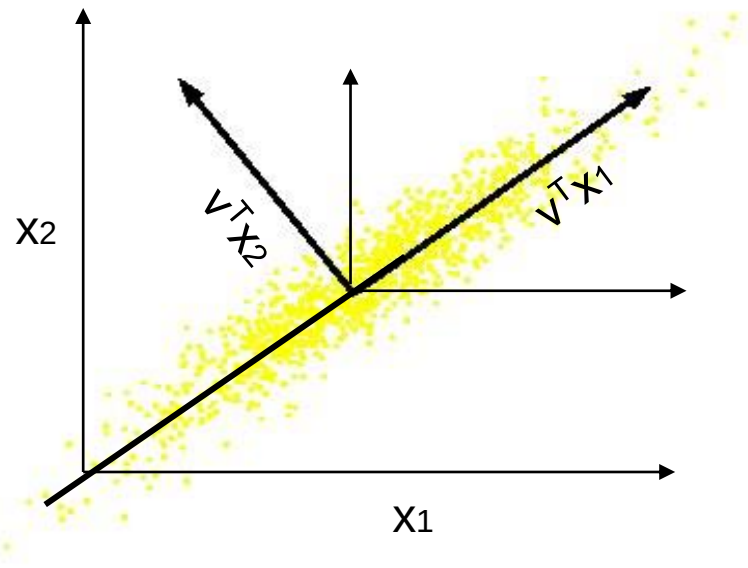
$$(\mathbf{X}\mathbf{X}^T - \lambda\mathbf{I})\mathbf{v} = 0$$

For symmetric matrices, eigenvectors for distinct eigenvalues are orthogonal.

Principal Component Analysis (PCA)

So, the new axes are the eigenvectors of the matrix of sample correlations XX^T of the data, which capture the similarities of the original features based on how data samples project to the new axes.

Transformed features are uncorrelated.



- Geometrically: centering followed by rotation
 - Linear transformation

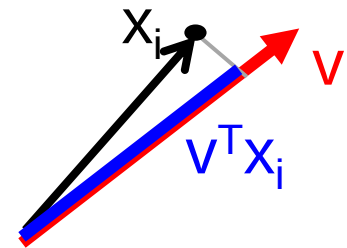
Another interpretation

Maximum Variance Subspace: PCA finds vectors $\mathbf{v}$ such that projections on to the vectors capture maximum variance in the data

$$\frac{1}{n} \sum_{i=1}^n (\mathbf{v}^T \mathbf{x}_i)^2 = \mathbf{v}^T \mathbf{X} \mathbf{X}^T \mathbf{v}$$

Minimum Reconstruction Error: PCA finds vectors $\mathbf{v}$ such that projection on to the vectors yields minimum MSE reconstruction

$$\frac{1}{n} \sum_{i=1}^n \|\mathbf{x}_i - (\mathbf{v}^T \mathbf{x}_i) \mathbf{v}\|^2$$

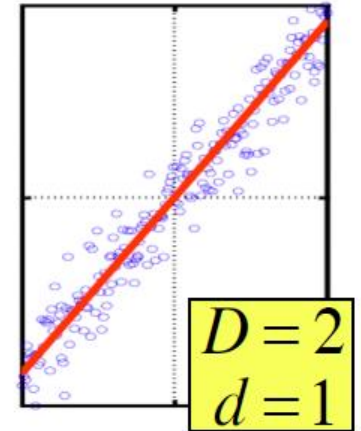


Dimensionality Reduction using PCA

The eigenvalue λ denotes the amount of variability captured along that dimension.

Zero eigenvalues indicate no variability along those directions => data lies exactly on a linear subspace

Only keep data projections onto principal components with non-zero eigenvalues, say $v_1, \dots, v_d$ where $d = \text{rank}(XX^T)$



Original Representation
data point

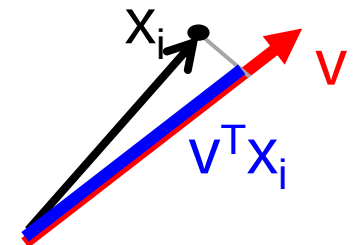
$$x_i = [x_i^1, x_i^2, \dots, x_i^D]$$

(D-dimensional vector)

Transformed representation
projections

$$[v_1^T x_i, v_2^T x_i, \dots, v_d^T x_i]$$

(d-dimensional vector)

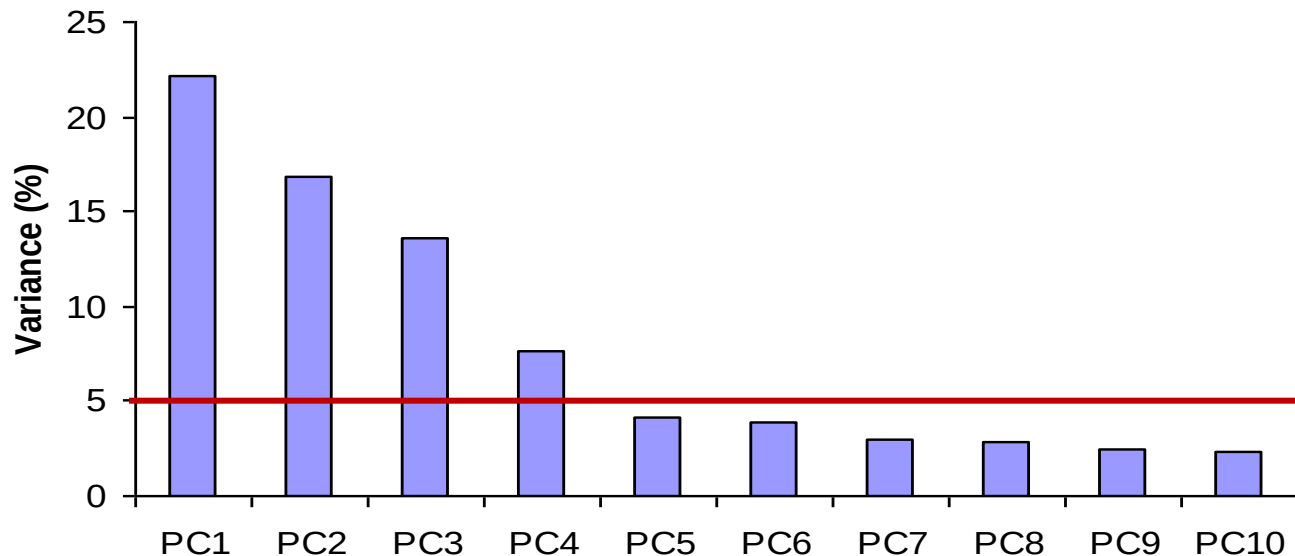


Dimensionality Reduction using PCA

In high-dimensional problem, data usually lies near a linear subspace, as noise introduces small variability

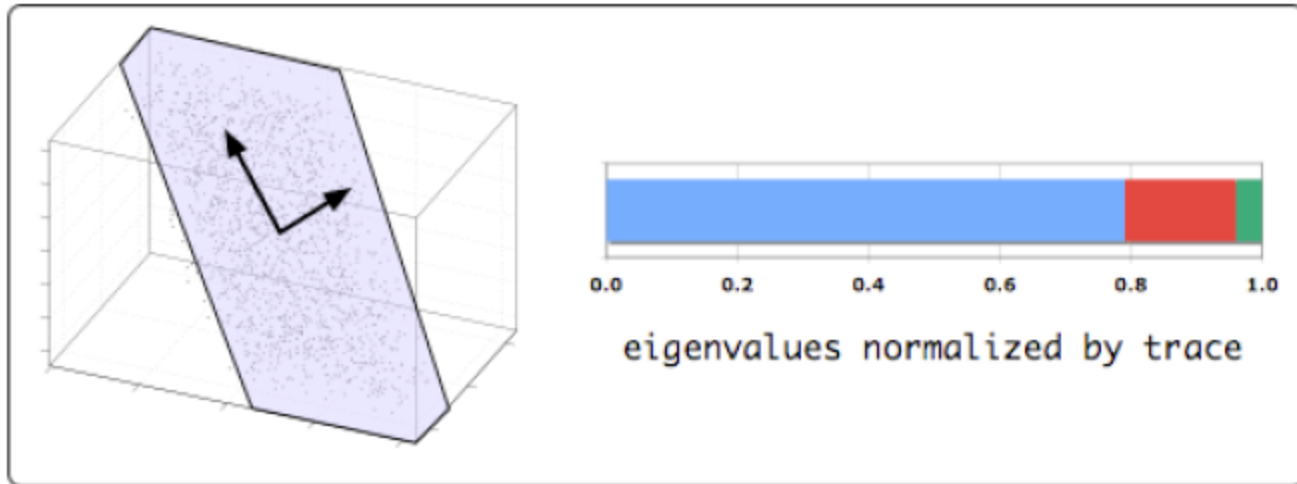
Only keep data projections onto principal components with **large** eigenvalues

Can *ignore* the components of lesser significance.



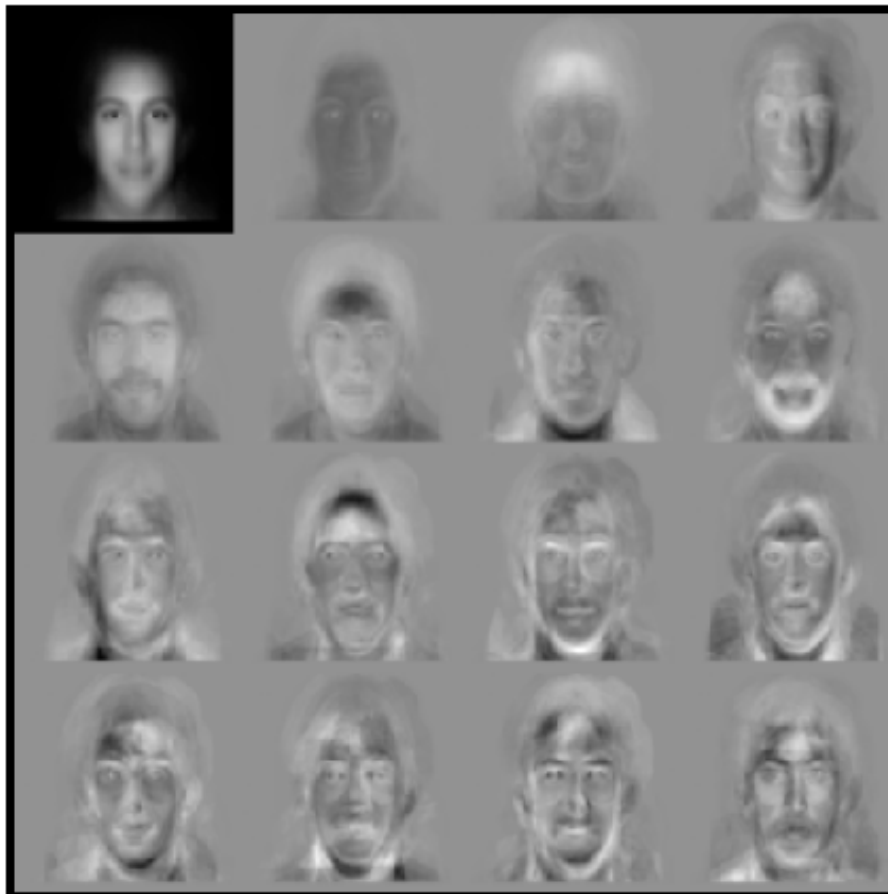
You might **lose some information**, but if the eigenvalues are small, you don't lose much

Example of PCA



Eigenvectors and eigenvalues of covariance matrix for $n=1600$ inputs in $d=3$ dimensions.

Example: faces



Eigenfaces
from 7562
images:

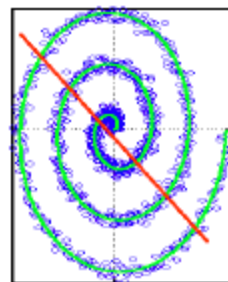
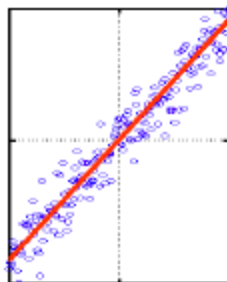
**top left image
is linear
combination
of rest.**

Sirovich & Kirby (1987)
Turk & Pentland (1991)

Properties of PCA

- **Strengths**

- Eigenvector method
- No tuning parameters
- Non-iterative
- No local optima



- **Weaknesses**

- Limited to second order statistics
- Limited to linear projections